Requirements Engineering

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Topics covered

- Requirements Engineering
- Requirements Elicitation
- Requirements Validation
- Requirements Management

Requirement Engineering

- **The goal of the requirements engineering process is to create and maintain a system requirements document**
- **Three sub-processes**
  - **Elicitation**
    - Discovering requirements (elicitation)
    - Converting the requirements into some standard form (specification)
  - **Validation**
    - Checking that the requirements
  - **Management**
    - Managing requirements changes

Requirements Elicitation

- **Requirements elicitation should find out about the application domain, what services the system should provide, the required performance of the system, hardware constraints, and so on**
  - **This may involve a variety of people (stake holders)**
    - Stakeholders are the people who will be affected by the system
  - **Stakeholders include end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc**

Difficulties with Eliciting Stakeholder Requirements

- There are several reasons for the difficulties
  - Stakeholders don't know what they really want
  - Stakeholders express requirements in their own terms and with implicit knowledge of their own work
  - Different stakeholders may have conflicting requirements
  - Political factors may influence the system requirements
  - The economic and business environment in which the analysis takes place is dynamic

Difficulties with Eliciting Stakeholder Requirements

- It is important to determine the following four elements of the project:
  - the purpose, aims and a basic understanding of the problem that requires a computer solution;
  - an understanding of the context and the environment in which the project will take place;
  - an understanding of the impact that the client expects the system to have;
  - an understanding of the motivation and expectations of the client

Requirements Elicitation Techniques

- Interviewing
- Scenarios
- Use-cases
- Ethnography
- Other techniques
  - Questionnaires
  - Focus Groups
  - Apprenticing
  - Brainstorming
  - Document Inspection

Interviewing

- **Formal or informal interviews with stakeholders**
  - **Puts questions to stakeholders**
  - **Requirements are derived from the answers**

- **Two types of interview**
  - **Closed interviews**
    - The stakeholders answer a pre-defined set of questions
  - **Open interviews**
    - There is no pre-defined agenda
    - Explores a range of issues with system stakeholders and hence develops a better understanding of their needs

Sample Questions for Interview

1. What is the new system intended to achieve? What is it intended to replace? What sort of system would you like?
2. Who will the users be? Who will the operators and maintainers of the system be?
3. Is there an existing system which is similar? What is wrong with it? How can it be improved? Is there are manual or paper system in place? Where can it be improved? Why have you decided to automate? What savings or economies do you expect?
4. What will the real benefit of the new system be to you?
5. What sort of environment will the system be used in?

Sample Questions for Interview

6. Can you provide examples of inputs and outputs for the system? Can you provide samples of reports or data for the existing manuals system?
7. What standards does your company use? What standards would affect this project?
8. What special properties do you expect this system to have? Should it be:
   - safe
   - reliable
   - high performance or real-time
   - secure
   - have high levels of usability

Interviews in Practice

- Interviews are normally a mix of closed and open interviews
  - Most interviews require some questions to get started and to keep the interview focused on the system to be developed
  - The answers may lead to other issues that are discussed in a less structured way

- Interviews are good for getting an overall understanding of:
  - What stakeholders do
  - How they might interact with the system
  - The difficulties that they face with current systems

Effective Interviewers

- Interviewers should be open-minded, avoid preconceived ideas about the requirements, and willing to listen to stakeholders
  - If the stakeholder comes up with surprising requirements, they are willing to change their mind about the system
- They should prompt the interviewee to start with a question, a requirements proposal, or by suggesting working together on a prototype system
  - Saying to people 'tell me what you want' is unlikely to result in useful information
  - Most people find it much easier to talk in a defined context rather than in general terms

Scenarios

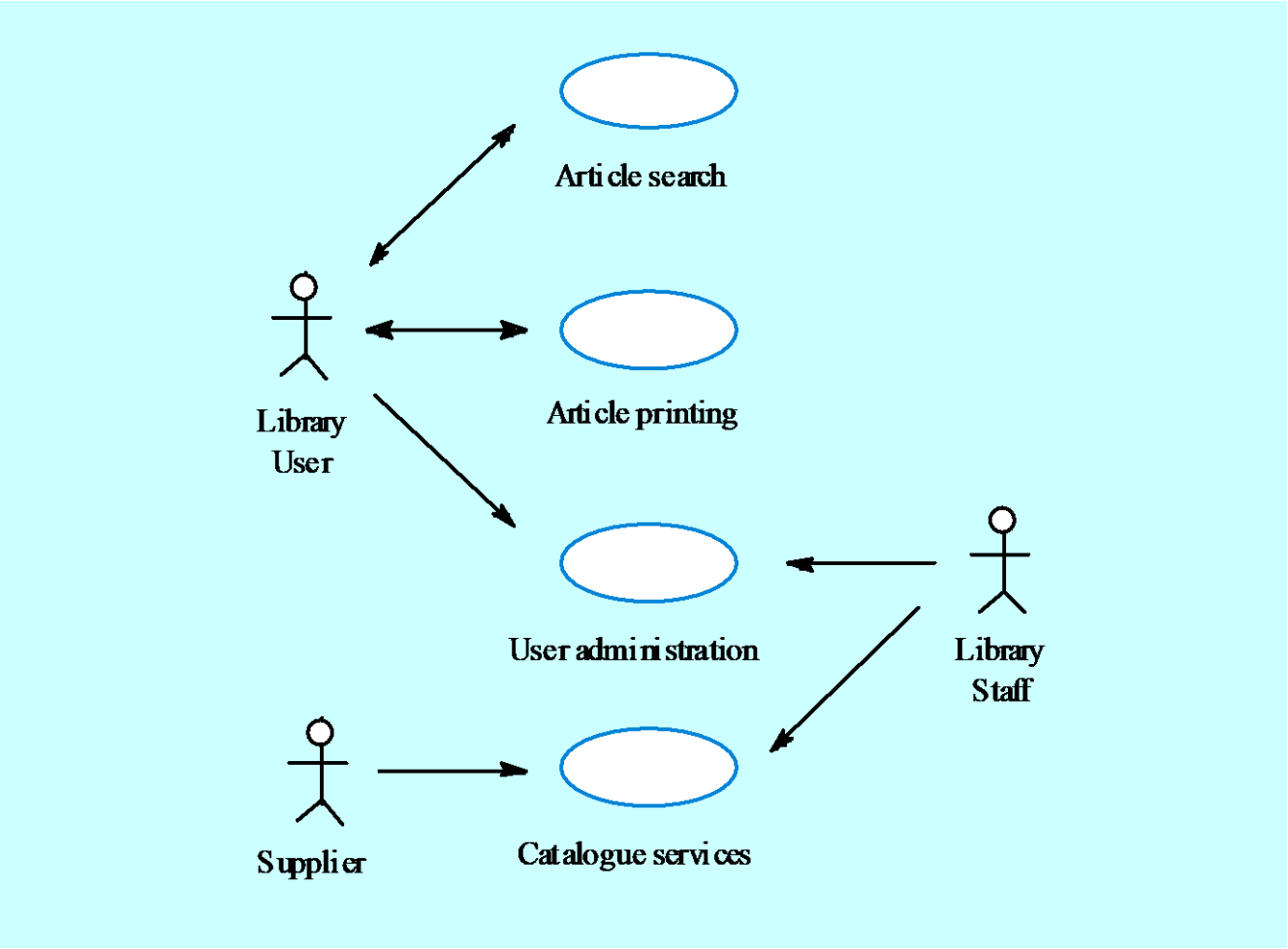
- Scenarios are real-life examples of how a system can be used
  - People usually find it easier to relate real-life examples than to abstract descriptions
  - Scenarios can be particularly useful for adding detail to an outline requirements

- A scenario may include;
  - A description of the starting situation
  - A description of the normal flow of events
  - A description of what can go wrong and how that is handled
  - Information about other concurrent activities
  - A description of the state when the scenario finishes

Use-Cases

- Scenarios may be written as text, supplemented by diagrams, screen shots, and so on
  - A more structured approach such as event scenarios or use-cases may be adopted
- Use-cases are a scenario based technique which were first introduced in the Objectory method (Jacobson, 1993)
  - They have now become a fundamental feature of the UML
- Scenarios and use-cases are not effective for eliciting constraints or high-level business and non-functional requirements, or for discovering domain requirements
  - Because they focus on interactions

Use-Case Example



Ethnography

- Ethnography is an observational technique that can be used to understand social and organizational requirements
  - An analyst immerses him or herself in the working environment where the system will be used
  - He or she observes the day-to-day work
  - The ethnography helps analysts discover implicit system requirements that reflect the actual rather than the formal processes in which people are involved

Ethnography

- People often find it very difficult to articulate details of their work because it is second nature to them
  - Some factors that affect the work but that are not obvious to individuals may only become clear when noticed by an unbiased observer

- Ethnographic studies can reveal critical process details that are often missed by other requirements elicitation techniques
  - However, ethnography is not a complete approach and should be used to complement other approaches, such as use-case analysis

Apprenticing

- The requirements engineer actually conducts the tasks that users of the system will be carrying out
  - This has the added benefit that the requirements analyst gains critical insight into the tasks required of the new system

Questionnaires

- Questionnaires can gather large amount of data quickly, can be precisely targeted and are relatively inexpensive to run

- Problems with questionnaires
  - Designing questionnaires can be difficult, questions can be ambiguous and the lack of direct contact prevents clarification of issues
  - A poor response rate
  - The data can remain untested unless good follow-up methods are employed
  - A danger that the respondents may be careless in filling in questionnaires

Brainstorming

- Brainstorming can be an effective way to generate lots of ideas in a short space of time
  - It works by having a number of people in a relaxed informal environment
  - The statement or problem for which ideas are sought is stated and then the people in the session simply shout out ideas until the time limit is reached
  - All of the ideas are captured and after the meeting these ideas are sifted to determine which idea(s) best solves the problem
  - There are rules to a brainstorming session, for example, no criticism of ideas is allowed and no discussion of ideas is allowed

- Brainstorming is good for creative input into a requirements session, perhaps when something new is being sought or there is a difficult problem to overcome
  - Brainstorming is not appropriate for any kind of analysis or for decision making

Document Inspection

- Inspecting documents is a good means of gathering requirements where existing systems have been formalized
  - Documents are a good source of information for checking facts from other sources as they do not rely on staff cooperation
- Informal systems and processes, or those systems and processes that are not documented are typically missed
  - Further, there is a problem with documents eventually becoming dated and so there is a need for the requirements analyst to ensure that documents are current and "*live*"!
- Document inspection can be time consuming so there is often a need to be selective
  - Also, data may not be in an immediately usable form and there is usually no information on attitudes or opinions

Topics covered

- Requirement Engineering
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Requirements Validation

- Requirements validation is concerned with showing that the requirements actually define the system that the customer wants
  - Requirements validation overlaps analysis in that it is concerned with finding problems with the requirements
  - Requirements validation is important because errors in a requirements document can lead to extensive rework costs when they are discovered during development or after the system is in service

Requirements Checking

- **During the requirements validation process, checks should be carried out on the requirements**

- **These checks include:**
  - **Validity checks**
    - Does the system provide the functions which best support the customer's needs?
  - **Consistency checks**
    - Are there any requirements conflicts?
  - **Completeness checks**
    - Are all functions required by the customer included?
  - **Realism checks**
    - Can the requirements be implemented given available budget and technology
  - **Verifiability**
    - Can the requirements be tested?

Requirements Validation Techniques

- **Requirements reviews**
  - Systematic manual analysis of the requirements
- **Prototyping**
  - Using an executable model of the system to check requirements
- **Test-case generation**
  - If the tests for the requirements are devised as part of the validation process, this often reveals requirements problems
  - If a test is difficult or impossible to design, this usually means that the requirements will be difficult to implement and should be reconsidered

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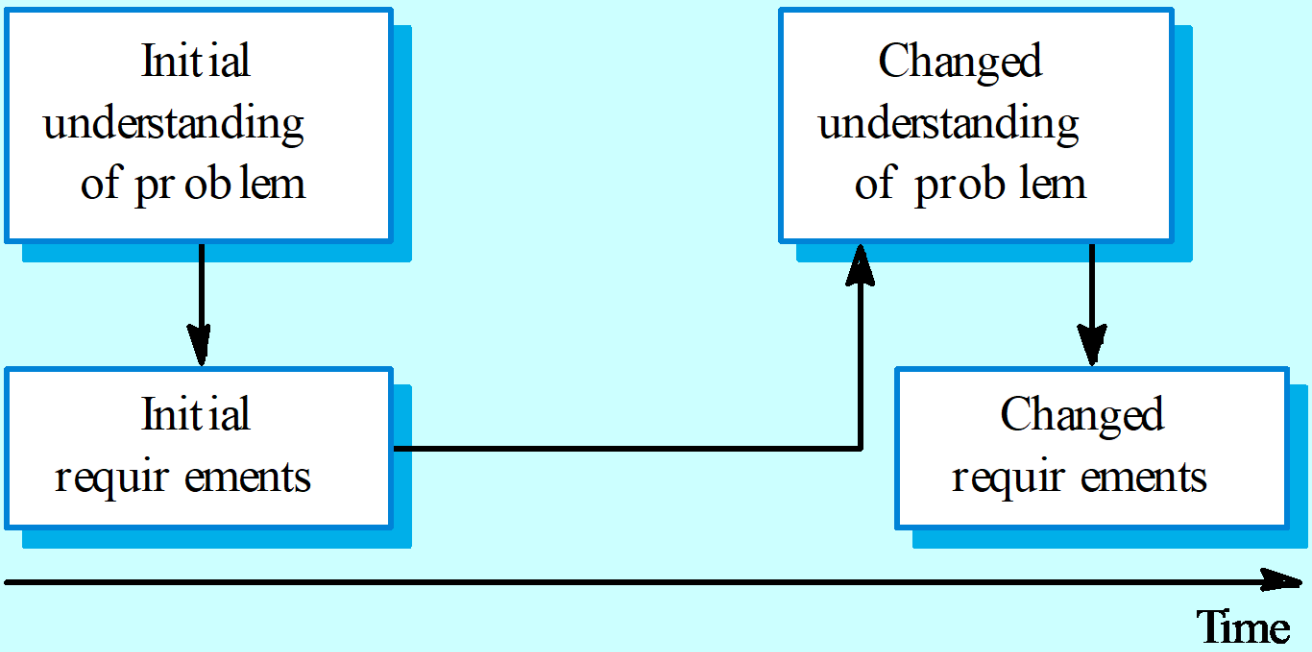
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Requirements Management

- The requirements for large systems are always changing
  - Because the problem cannot be fully defined, the software requirements are bound to be incomplete
  - The stakeholders' understanding of the problem is constantly changing
  - Once a system has been installed, new requirements inevitably emerge

- Requirements management is the process of understanding and controlling changes to system requirements
  - You need to keep track of individual requirements and maintain links between dependent requirements so that you can assess the impact of requirements changes

Requirements Evolution



Enduring and Volatile Requirements

- **From an evolution perspective, requirements fall into two classes**

- **Enduring requirements**
  - **Relatively stable requirements derived from the core activity of the customer organisation**
    - E.g. A hospital will always have doctors, nurses, etc.
    - May be derived from domain models

- **Volatile requirements**
  - **Requirements which change during development or when the system is in use**
    - E.g. In a hospital, requirements derived from health-care policy

Traceability

- **Traceability is the property of a requirements specification that reflects the ease of finding related requirements**
  - **Source traceability**
    - Links from requirements to stakeholders who proposed these requirements
  - **Requirements traceability**
    - Links between dependent requirements
  - **Design traceability**
    - Links from the requirements to the design

Traceability matrices

- **Traceability information is often represented using traceability matrices**
  - **Each requirement is entered in a row and in a column in the matrix**
  - **A ‘D’ in the row/column intersection illustrates that the requirement in the row depends on the requirement named in the column**
  - **An ‘R’ means that there is some other, weaker relationship between the requirements**
    - For example, they may both define the requirements for parts of the same subsystem

A traceability matrix

Req. id	1.1	1.2	1.3	2.1	2.2	2.3	3.1	3.2
1.1		D	R					
1.2			D			D		D
1.3	R			R				
2.1			R		D			D
2.2								D
2.3		R		D				
3.1								R
3.2							R	


Q & A

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